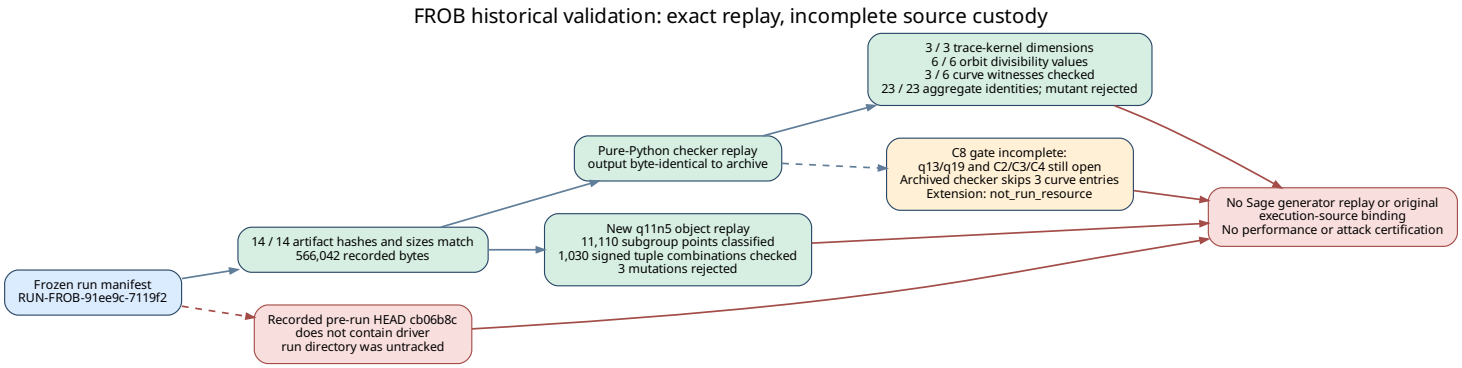
Historical FROB checker replay and source custody

Validation audit, 22 October 2026

Scope. This checks frozen run `RUN-FROB-91ee9c-7119f2` at source tree `e83bb2f14414c9669360816aeee5876dfcfd4384`. It is an integrity audit, not a new FROB measurement or a research verdict. The run directory was copied before executing its pure-Python checker; frozen bytes were not edited.

Check	Result	Limit
Archived artifact hashes and sizes	14/14 match; 566,042 bytes	Storage integrity only
Checker output	Byte-identical replay	Same archived checker source
Trace-kernel dimension; orbit divisibility	3/3; 6/6	Recorded core cells
Curve point/order/Frobenius witnesses	3/6 checked	Other three skipped for size
Aggregate rational identities	23/23; mutant rejected	Counts treated as inputs
New q11n5 object-arm replay	11,110 points; 1,030 tuples	Both eligible curves; 3 mutants rejected
Remaining C8 work	Open	q13/q19 object arms; C2/C3/C4
Extension cell	`not_run_resource`	540-second historical checkpoint



Source custody. The manifest records pre-run HEAD `cb06b8c20c2d6559041c45c4a4b29dd7f132fdf5` and says the run directory was untracked at execution start. The driver is absent from that commit. Current-schema `code.commit`, `code.command`, and `environment` fields are absent, although separate command and environment artifacts are hash-pinned. Their presence does not bind the executed driver to the recorded HEAD.

Boundary and next check. The archived checker's `tuple_sum_reverifications` label overstates its code: its three `ok` entries check curve witnesses, not tuple sums or control arms. The new q11n5 replay verifies its object arm with the archived checker's field and curve primitives; the other cells and all controls remain open. Thus C8 and the success gate are unestablished. Neither replay binds the original Sage execution source or proves attack performance. Recover a contemporaneous source snapshot or run a fresh source-bound replay under the frozen controls. The Coordinator must adjudicate the original record without assigning the pre-run HEAD as a false execution-source commit. Receipts: `audit.json`, `split-object-replay.json`.